

From Legal PDFs to Verifiable Answers

Answer quality, page-level grounding, and latency across 100 legal questions

github.com/SteinVR/Agentic-RAG-Challenge
Trends in Machine Learning II · Universität Trier

01 HYPOTHESIS

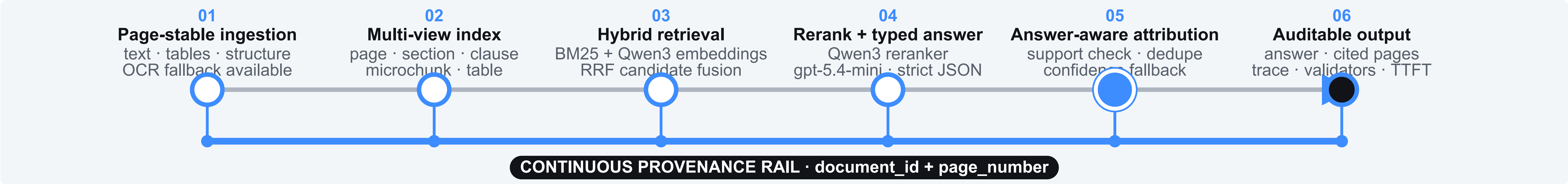
RESEARCH QUESTION

Can a page-traceable Legal RAG pipeline balance answer quality, grounding, valid output, and latency?

- H1 Targets: answer $\geq .90$ · grounding $\geq .85$ · 100% valid · 95/95 hits · latency $\geq .90$.
- H2 Attribution should raise precision and reduce cited pages while preserving hit coverage.

02 METHODOLOGY

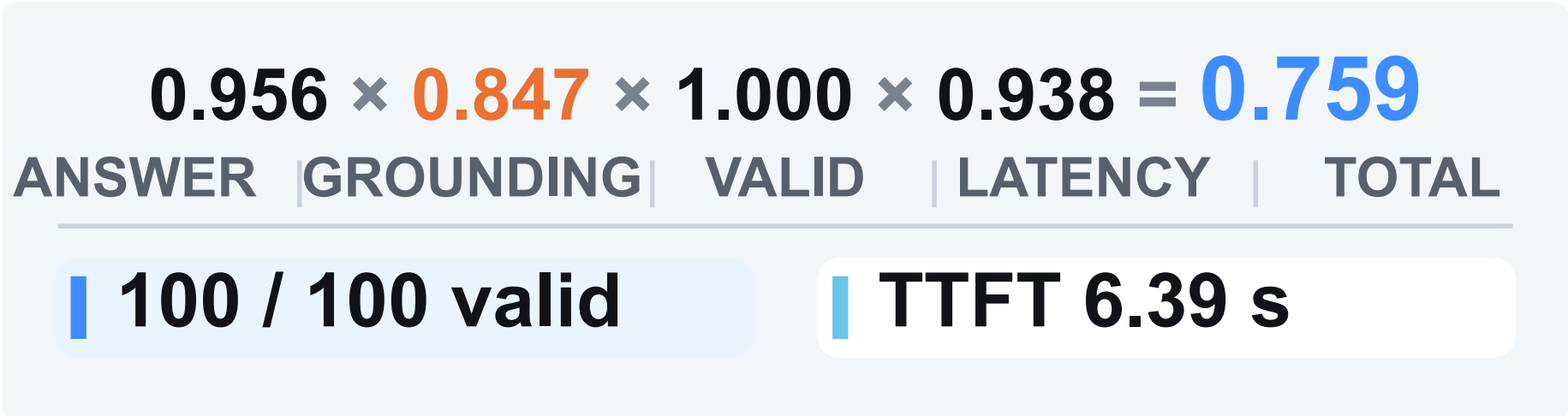
30 public legal PDFs · 590 pages · 100 questions · 95 page-annotated · 32 boolean · 30 free text · 17 number · 15 name · 5 names · 1 date · paired stage comparison · 20,000 bootstrap resamples



03 RESULTS

Strong answers, multiplicative bottlenecks

End-to-end benchmark profile · all 100 answers



H1 is only partly supported: benchmark grounding missed the retrospective 0.85 threshold by 0.003.

Do not merge the two grounding views: benchmark grounding is 0.847 on 100 answers; macro page F β is 0.892 on 95 annotated questions.

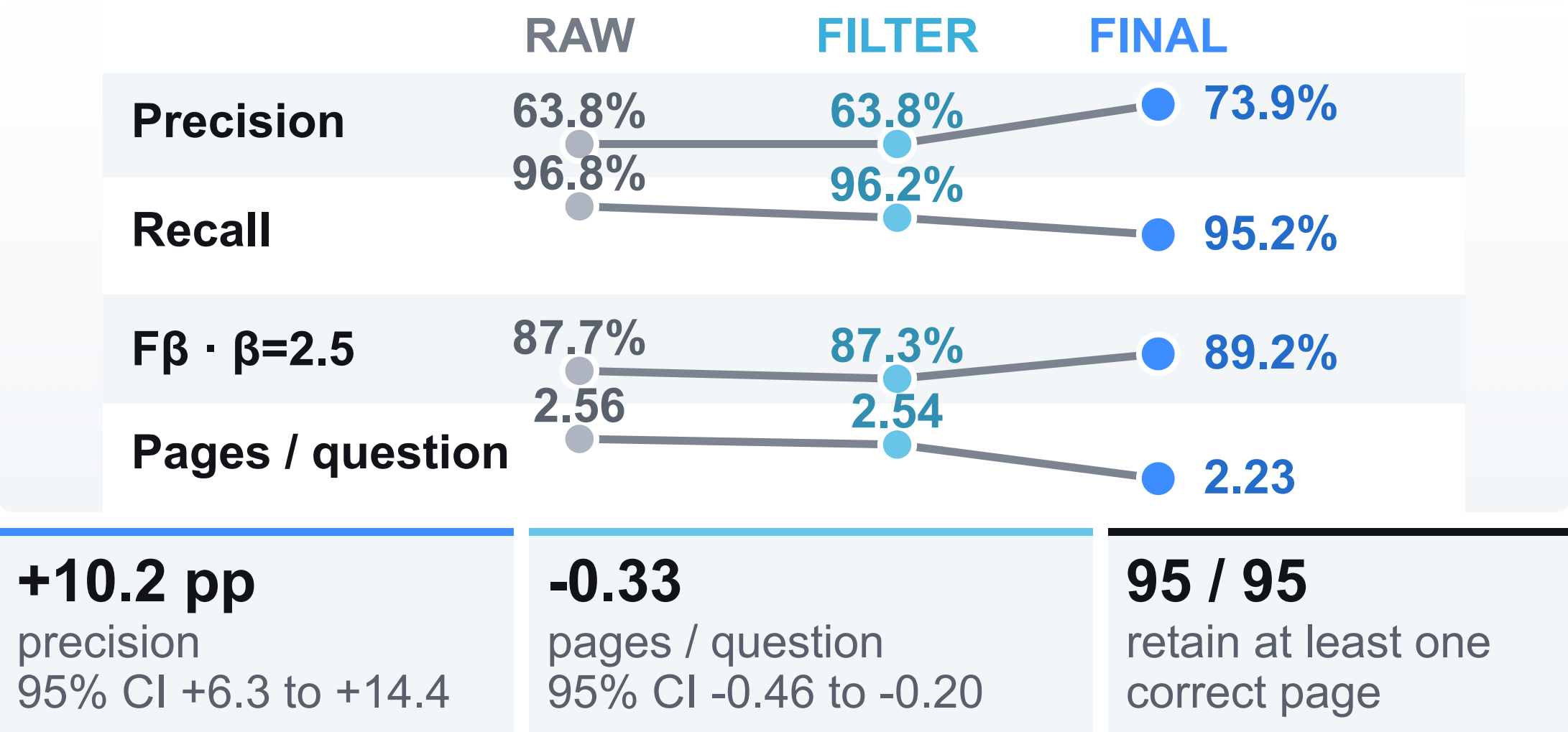
DECISION

Retrieve broadly. Attribute explicitly. Narrow conservatively when support spans pages or documents.

Sequential states of the same answers - not three independent systems.

Attribution removes excess pages without complete misses

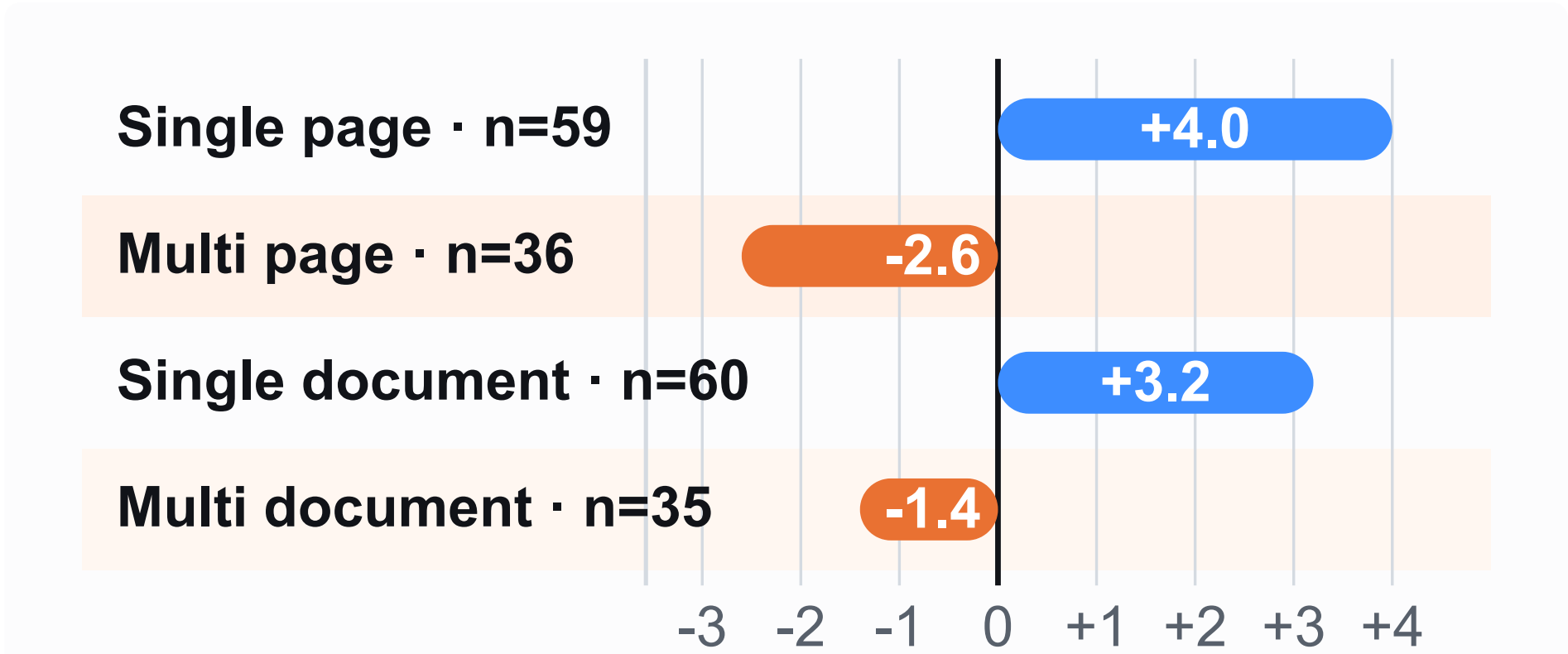
Sequential states of the same 95 annotated answers



H2 is supported in its primary effects. Recall fell 1.6 pp; F β rose 1.5 pp, but its paired interval spans zero (-0.3 to +3.1 pp).

Narrowing helps local evidence, hurts distributed evidence

Final attribution minus raw retrieval · macro F β delta (pp)



Distributed support is the failure boundary: the selector becomes too aggressive when evidence spans pages or documents.

Exploratory split, not a pre-registered causal test. The pipeline stages are architecture, not a component ablation.

LIMITATIONS

- One saved run; no repeated seeds or end-to-end uncertainty estimate.
- Retrospective thresholds; no full component ablation or independent baseline.
- One DIFC corpus; TTFT only; OCR fallback was not separately exercised.